25900911 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCPHY100 - FOUNDATIONS OF PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **7** Questions

Each Question carries **2** marks

1. Find the magnitude of the vector **A** = 6i - 3j + 2k . [A] / [CO1]
2. Give examples of significant figures. [K] / [CO1]
3. Which quantity is obtained from the area of velocity time graph? Explain. [U] / [CO3]
4. What is relative velocity? [K] / [CO3]
5. Explain why mass is considered a scalar quantity, while weight is a vector quantity. [U] / [CO3]
6. Define conservative force. Give an example for conservative force. [U] / [CO4]
7. What are stable equilibrium and unstable equilibrium positions? [K] / [CO4]
8. Describe the weight of a body? Is it a constant? [U] / [CO3]

9. What is the differnce between input and raw_input? [U] / [CO5]

10. What is a comment in python programming? [K] / [CO5]

[7x2 = 14]

Part B

Short Essay Questions

Answer any **4** Questions

Each Question carries **6** marks

11. Define projectile motion. What are the assumptions made in the study of projectile motion? [K] / [CO3]

12. A particle is observed initially at position (2 , 3.5). Three seconds later it is observed at (-3 , 5.5), where the distances are expressed in meter. What is the particle's average velocity during this time? [A] / [CO3]

13. Derive the expression for terminal speed in a fluid medium. [U] / [CO3]

14. What is the difference between positive work, negative work, and zero work? [U] / [CO4]

15. What are the different forms of energy? [U] / [CO4]

16. Write a python programme to sum all the elements in the List [10,4,5,7,8,9,11,12]. [A] / [CO5]

[4x6 = 24]

Part C

Essay Questions

Answer any **1** Question

Each Question carries **12** marks

17. Discuss in detail, the position - time graph. Describe how average velocity can be calculated. Draw the x - t graph for a body moving with constant velocity. [A] / [CO2]

18. Explain elastic potential energy and arrive at an expression for it. Also discuss the transformation of energies when a diver jumps off a high board into a swimming pool, assume that air resitance is present. [A] / [CO4]

[1x12 = 12]